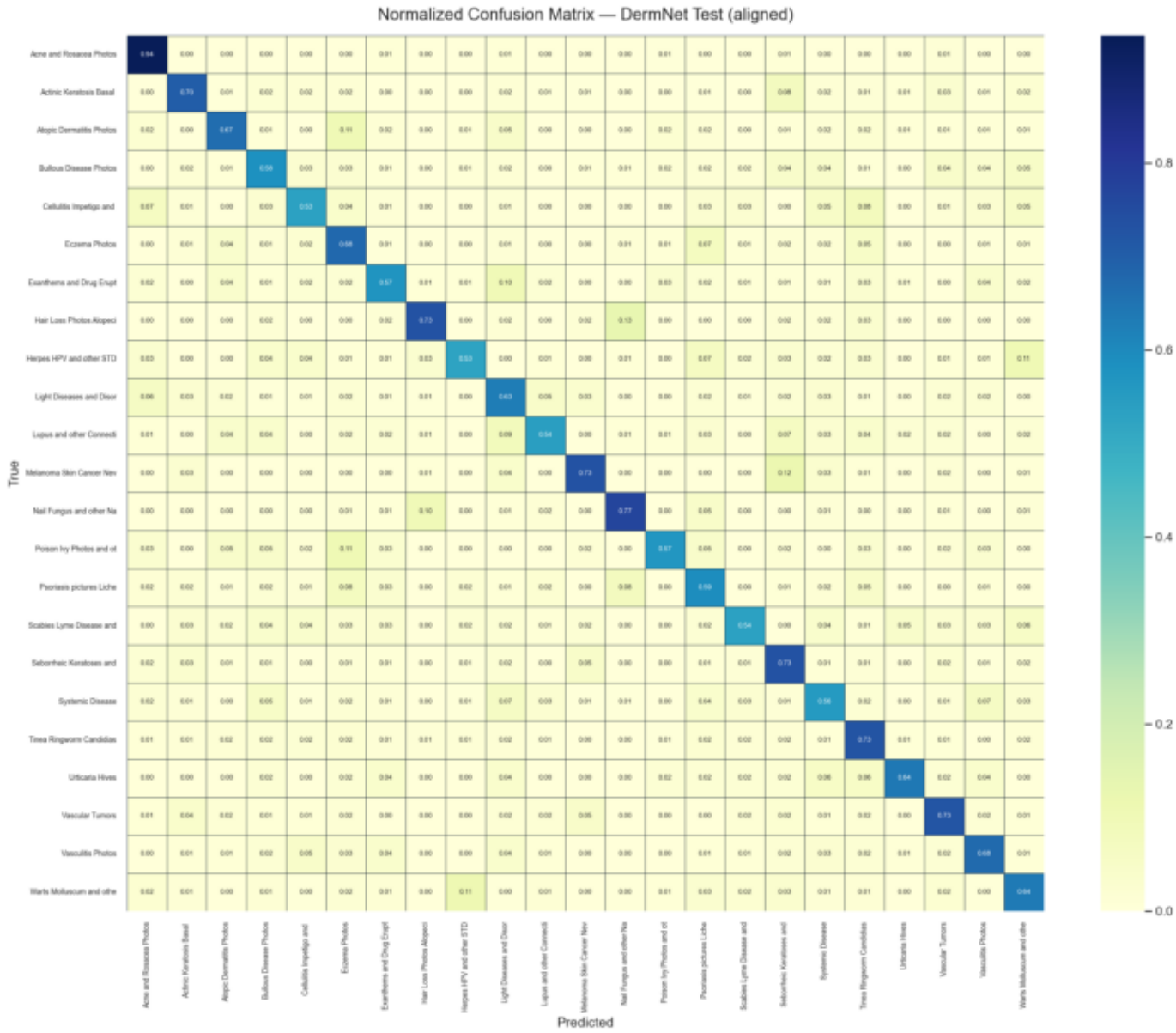
DermAI — Conference Performance Report

EfficientNet-B4 skin-lesion classifier • DermNet held-out test split • evaluation with empirically aligned label mapping

Metric	Value
Top-1 Accuracy (%)	68.06596701649175
Top-3 Accuracy (%)	86.2568715642179
Precision (macro, %)	64.14926244898052
Recall (macro, %)	65.30059954993611
F1-score (macro, %)	64.48432050120597
F1-score (weighted, %)	68.1181796666161
ROC-AUC (macro)	0.9626335218339672
Brier score	0.47284618842485143
ECE (Expected Calibration Error)	0.16133175936268307
MCE (Max Calibration Error)	0.29763359748400176
Mean confidence (%)	84.19915008544922
Mean conf. on correct (%)	91.56695556640625
Mean conf. on incorrect (%)	68.4949722290039
Test images	4002
Classes	23
Inference/sec (CPU)	—

Confidence-Level Interpretation (for conference narrative)

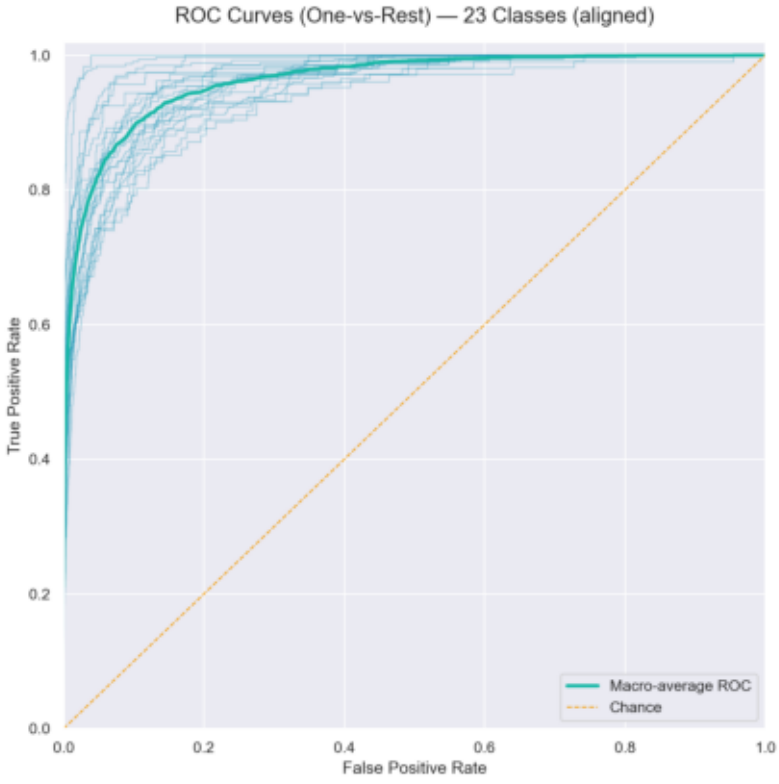
• Correct predictions are markedly more confident: 91.6% vs 68.5% on errors (spread 23.1 pts).
• Above 70% confidence, accuracy reaches 79.5% (76.1% of all predictions).
• ECE 0.161 / MCE 0.298 show the model is overconfident — a temperature-scaling candidate on the Pareto frontier for the paper.
• 0.0% of test predictions fall under the app's 5% healthy-override gate and are auto-labelled 'Healthy Skin'.


Normalized Confusion Matrix — DermNet Test (aligned)
True
Predicted
Acne and Rosacea Photos
Actinic Keratosis Basal
Atopic Dermatitis Photos
Bullous Disease Photos
Cellulitis Impetigo and
Eczema Photos
Exanthems and Drug Erupt
Hair Loss Photos Alopeci
Herpes HPV and other STD
Light Diseases and Disor
Lupus and other Connecti
Melanoma Skin Cancer Nev
Nail Fungus and other Na
Poison Ivy Photos and ot
Psoriasis pictures Liche
Scabies Lyme Disease and
Seborrheic Keratoses and
Systemic Disease
Tinea Ringworm Candidias
Urticaria Hives
Vascular Tumors
Vasculitis Photos
Warts Molluscum and othe
0.8
0.6
0.4
0.2
0.0

Per-Class Metrics (DermNet Test, sorted by F1)

class	model_output_index	support	precision	recall	f1_score	roc_auc	pr_auc	mean_confidence_pct	class_accuracy_pct
Acne and Rosacea	0	312	0.85	0.94	0.89	0.99	0.97	96.25	93.59
Nail Fungus and ot	13	261	0.81	0.77	0.79	0.99	0.89	93.87	77.01
Actinic Keratosis B	1	288	0.8	0.7	0.75	0.97	0.82	83.73	70.49
Tinea Ringworm C	19	325	0.74	0.73	0.73	0.96	0.8	85.31	72.92
Seborrheic Keratos	17	343	0.73	0.73	0.73	0.97	0.81	82.97	73.18
Melanoma Skin Ca	12	116	0.7	0.73	0.72	0.99	0.78	88.34	73.28
Warts Molluscum a	23	272	0.73	0.64	0.68	0.96	0.77	85.48	63.97
Eczema Photos	5	309	0.68	0.68	0.68	0.96	0.74	83.77	67.64
Vascular Tumors	21	121	0.61	0.73	0.66	0.98	0.79	85.47	72.73
Atopic Dermatitis P	2	123	0.65	0.67	0.66	0.97	0.74	84.07	66.67
Urticaria Hives	20	53	0.67	0.64	0.65	0.97	0.73	83.64	64.15
Psoriasis pictures L	15	352	0.7	0.59	0.64	0.94	0.71	81.32	59.38
Vasculitis Photos	22	105	0.61	0.68	0.64	0.97	0.74	83.14	67.62
Hair Loss Photos A	7	60	0.54	0.73	0.62	0.99	0.64	81.14	73.33
Poison Ivy Photos	14	65	0.66	0.57	0.61	0.97	0.66	78.99	56.92
Scabies Lyme Dise	16	108	0.62	0.54	0.57	0.94	0.63	76.36	53.7
Light Diseases and	10	143	0.52	0.63	0.57	0.95	0.67	81.3	62.94
Lupus and other C	11	105	0.59	0.54	0.56	0.94	0.6	78.0	54.29
Systemic Disease	18	152	0.56	0.56	0.56	0.96	0.65	74.85	55.92
Exanthems and Dr	6	101	0.53	0.57	0.55	0.93	0.59	82.45	57.43
Bullous Disease Pi	3	113	0.5	0.58	0.54	0.95	0.62	78.08	58.41
Herpes HPV and o	9	102	0.52	0.53	0.53	0.95	0.57	80.72	52.94
Cellulitis Impetigo a	4	73	0.45	0.53	0.49	0.94	0.5	74.75	53.42

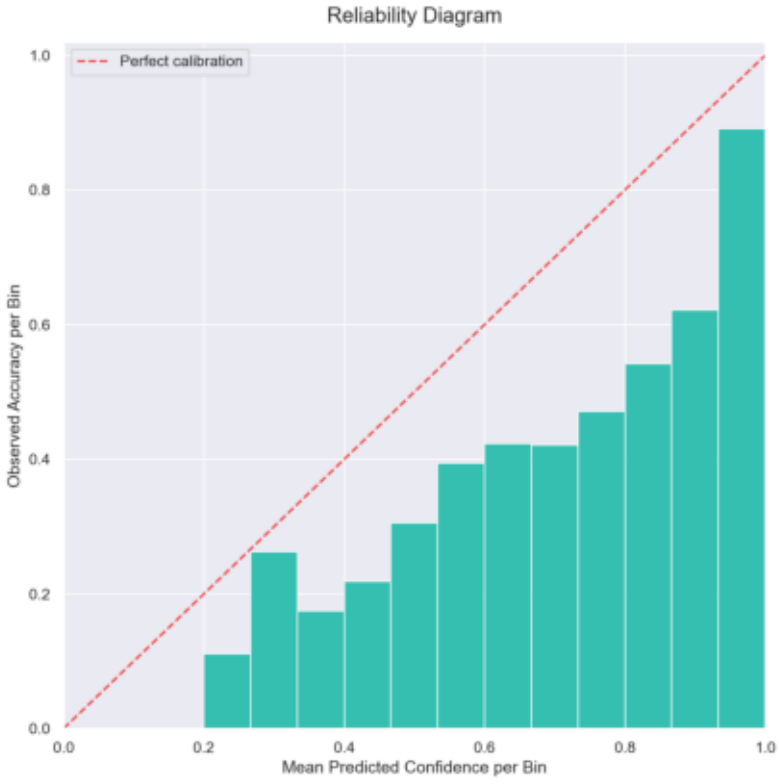
Discriminative Performance & Data Coverage





Confidence Analysis & Calibration





Reliability / Calibration Bins (15 bins)

confidence_bin_low	confidence_bin_high	samples	fraction_of_total_pct	bin_accuracy_pct	bin_mean_confidence_pct
0.0	0.07	0.0	0.0	0.0	0.0
0.07	0.13	0.0	0.0	0.0	0.0
0.13	0.2	2.0	0.05	0.0	17.4
0.2	0.27	27.0	0.67	11.11	24.28
0.27	0.33	61.0	1.52	26.23	30.44
0.33	0.4	97.0	2.42	17.53	36.89
0.4	0.47	142.0	3.55	21.83	43.15
0.47	0.53	177.0	4.42	30.51	50.15
0.53	0.6	165.0	4.12	39.39	56.8
0.6	0.67	182.0	4.55	42.31	63.34
0.67	0.73	185.0	4.62	42.16	69.73
0.73	0.8	208.0	5.2	47.12	76.88
0.8	0.87	251.0	6.27	54.18	83.34
0.87	0.93	309.0	7.72	62.14	90.29
0.93	1.0	2196.0	54.87	89.12	98.92



Notes for the conference manuscript

- Label alignment: the checkpoint authors exported 24 output heads; DermNet has 23 classes. Model-output indices 0-7 map 1:1; indices 8-22 map to class indices 9-23 (a phantom head sits at model index 8). All metrics here are computed under that empirically verified alignment (see alignment_map.csv). A mismatch like this silently caps accuracy otherwise.
- The 5% confidence gate in the live app is effectively a reject/abstain rule; consider reporting accuracy-with-rejection, which is what the deployed system actually delivers.
- Recommend reporting per-class F1 rather than macro alone given class imbalance (see support).